

Coding & Solution

Date	1 July 2026
Team ID	xxxxxx
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Repository Link / URL	https://github.com/manankochar1720/Gen-ai-Google-Gemini-Powered-Learning-Assistant
Programming Language(s)	Python, JavaScript, HTML, CSS
Framework(s) Used	FastAPI, Streamlit, Uvicorn
Key Features Implemented	1. Concept Explainer: personalized AI explanation guides by difficulty and style. 2. AI Tutor Chat: interactive chatbot with focus-topic query tracking. 3. Practice Quiz: dynamic quiz engine with multiple-choice questions and instant feedback. 4. Learning Chronicle: session history logs with offline PDF downloads. 5. Ambient Dark Mode: dark shade dashboard styling with floating ambient neon blobs. 6. Interactive Header controls: popover panels for alerts/notifications and account details.
Pending / Incomplete Features	User authentication, database integration, cloud deployment
Setup / Run Instructions	1. Clone repository. 2. Create virtual environment. 3. Install dependencies using <code>pip install -r requirements.txt</code>. 4. Run backend using <code>python -m uvicorn app.main:app --reload</code>. 5. Run frontend using <code>streamlit run frontend/streamlit_app.py</code>.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate frontend, backend, service, and testing layers. The application was successfully tested using PyTest, and all test cases passed successfully (5/5 tests passed). The project is fully functional and ready for further enhancements and deployment.
